

RYAN WEBSTER

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EDUCATION

Bachelor of Science, Mechanical Engineering | University of Massachusetts Lowell | GPA 3.41 Sept 2020- May 2025

EMPLOYMENT

R&D Technician | **FORMLABS**, Somerville MA Nov 2025 - Present

- Developed test procedures to evaluate the reliability of optical subsystems for an SLA 3D printer. Down selected a new LCD adhesive layer by testing for impact resistance, and degradation of spectral transmittance over time.
- Debugged system-level motion control issues, and automated printer calibration routines using Linux/Python.
- Built prototypes of next-generation 3D printers, reworked electronic components/sheet metal parts in the machine shop, and troubleshooted build issues with the manufacturing engineering team.

R&D Engineering Intern | **SPRYTE MEDICAL**, Bedford MA May 2024 - July 2025

- Simulated use-testing of catheter-based intraneural OCT imaging probes to troubleshoot the root causes of failure when deployed inside the brain, and compared multiple configurations to optimize image quality.
- Implemented automated camera vision systems into manufacturing line equipment, and programmed PLCs with physical feature detection tools to improve assembly accuracy and manufacturability.
- Oversaw hardware documentation control by creating and maintaining SolidWorks drawings. Wrote scripts using Windows PowerShell to automate large batches of revision changes for SOP documentation.

Mechanical Engineering Co-Op | **TERADYNE**, North Reading MA Dec 2022 - July 2023

- Designed a pneumatic test fixture to evaluate the lifecycle of a molded component with a high defect rate. Tested for mechanical failure criteria of the brackets to validate FEA results for the new material selection.
- Developed and revised hardware components to implement in field products for continuous improvement.
- Operated and maintained several FDM 3D printers to ensure smooth production of parts for rapid prototyping.

Machine Shop TA | **UMASS LOWELL**, Lowell MA Sept 2022 - Dec 2022

- Taught students how to operate manual mills, CNC mills, and CNC lathes to fabricate metal parts.

Plastics Engineering Research Assistant | **UMASS LOWELL**, Lowell MA May 2021 - Aug 2021

- Conducted research on plastic waste mitigation by utilizing FTIR Spectroscopy to analyze the absorption spectra of macroplastic beach litter samples, and identify the most frequently occurring polymers in ocean waste.
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SELECTED PROJECT EXPERIENCE

Riverhawk Racing (FSAE), Aerodynamics Team Sept 2022 - May 2024

- Designed bodywork components for an open wheel race car to improve aerodynamic performance, and performed CFD simulations to find drag coefficient, downforce, and analyze flow paths of air.
- Manufactured body panels from woven composite fabric, using an epoxy resin wet layup on convex foam molds.

Design, Build, Fly Sept 2024 - Dec 2024

- Designed the electronics rig for a remote-control airplane using the plane's physical characteristics to calculate the necessary electronic specs. Ensured the ESC, battery, motor, and propeller would generate sufficient thrust.

Applied Finite Element Analysis Jan 2025 – May 2025

- Utilized techniques such as: plotting localized stress/deformation, calculating failure criteria, mesh convergence, nodal analysis, and topology optimization to predict the mechanical behavior of structures under load.
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TECHNICAL SKILLS

DESIGN	SolidWorks / Onshape / Ansys Mechanical / Abaqus / Altair Hypermesh / Ansys Fluent / KiCAD
EQUIPMENT	3D Printing (FDM, SLA, SLS) / Machining (Manual, CNC) / Epilog Laser / Instron / Keyence KV-X
SCRIPTING	Python / MATLAB / Simulink / Powershell / Arduino / HTML / CSS